



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

COMMON HALF YEARLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours

BIOLOGY

[Max. Marks : 70]

Part - I (Bio-Botany) (Marks : 35)

Section - I

Note. (i) Answer All the questions.

8×1=8

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

- Which one of the following method are used to break the seed dormancy?
 - Scarification
 - Impaction
 - Stratification
 - All the above
- Assertion (A)** : Oxidative phosphorylation takes place during the electron transport chain in Mitochondria.
Reason(R) : Succinyl CoA is phosphorylated into succinic acid by substrate phosphorylation
 - A and R is correct, R is correct explanation of A
 - A and R is correct but R is not the correct explanation of A
 - A is correct but R is wrong
 - A and R is Wrong
- Usually, the monocotyledons do not increase their girth, because
 - They Possess activity dividing cambium
 - They do not possess actively of cambium
 - Ceases activity of cambium
 - All are correct
- The β - D Glucose units in Cellulose are linked together by
 - n - Acetyl side Chains
 - N - acetyl D- Glucosamine
 - 1 $\rightarrow$ 3 Linkage
 - β - (1,4) glycosidic linkage
- Phylogenetic classification is the most favoured classification because it reflects
 - Comparative Anatomy
 - Number of Flowers Produced
 - Comparative Cytology
 - Evolutionary relationships
- Identify correct Match**

1. Die back disease of citrus	- i) Mo	a) i - iii, 2 - ii, 3 - iv, 4 - i
2. Whip tail disease	- ii) Zn	b) i - iii, 2 - i, 3 - iv, 4 - ii
3. Brown heart of turnip	- iii) Cu	c) i - i, 2 - iii, 3 - ii, 4 - iv
4. Little leaf	- iv) B	d) i - iii, 2 - iv, 3 - ii, 4 - i
- In Gymnosperm endosperm is formed
 - At the time of Fertilization
 - Before Fertilization
 - After Fertilization
 - Along with the development of Embryo
- Identify the Mismatch Pair.**

a) Virion - Infective virus Particle	b) Viroid - Linear molecule of SSRNA
c) Virusoids - smaller circular RNAs	d) Prions - Proteinaceous infections Particles

SECTION - II

II. Answer any four of the following.

4×2=8

- Differentiate between Phylloclade and Cladode.
- Draw and label open and closed vascular bundle.
- Define Respiratory Quotient.
- What is Richmond lang effect?
- What is amber? which group of Plants produces Amber?
- Name the types of Mixed inflorescence.

SECTION - III

III. Answer any three questions. Question No. 19 is compulsory.

3×3=9

- Differentiate Cytokinesis in plant cells and animal cells.
- Write note on Zwitterion.
- Bring out the Importance of Growth Rings.
- How phosphorylase enzyme open the stomata in starch sugar interconversion theory?
- Give a brief account on Programmed cell Death (PCD)?

SECTION - IV

IV Answer all the questions.

2×5=10

- (a) Describe the Steps involved in Gram staining Method. (OR)
(b) Describe C3 Cycle (or) (Flow chart or Explanation Calvin Cycle)
- (a) What is the role of National Gardens in conserving Biodiversity - Discuss. (OR)
(b) Write the Physiological effects of Gibberellins.



PART – II (BIO-ZOOLOGY) (Max. Marks: 35)

SECTION – I

I. Choose the most appropriate answer.

8x1=8

1. What taxonomical aid gives comprehensive information about a taxon?
a) Taxonomic key b) Herbarium c) Flora d) Monograph
2. Non-Shivering thermogenesis neonate produces heat through
a) white fat b) Brown fat c) yellow fat d) colourless fat
3. Which of the following is correctly matched
a) Physalia - Portuguese man of war
b) Pennatula - sea fan
c) Adamsia - sea pen
d) Gorgonia - sea anemone
4. Glomerular filtrate contains
a) Blood without blood cells and proteins b) Plasma without sugar
c) Blood with proteins but without cells d) Blood without urea
5. Which of the following substances in Tobacco smoke damage the gas exchange system?
a) Carbon monoxide and Carcinogenes b) Carbon monoxide and nicotine
c) Carcinogens and tar d) nicotine and tar
6. Which of the following have an open circulator system?
a) Frog b) Earthworm c) Pigeon d) Cockroach
7. The pigment present in the muscle fibre is
a) sarcomere b) sarcoplasm c) myosin d) actin
8. Isinglass is used in
a) Preparation of wine b) cleaning of wines c) Distillation of wines d) preservation of wines

SECTION – II

II. Answer any Four of the following in one or two sentences.

4x2=8

9. What is Binomial Nomenclature?
10. Why flatworms are called acoelomates?
11. Some epithelia are Pseudostratified. What does this mean?
12. Why do you call Cockroach a pest?
13. Why is Pneumonia considered a dangerous disease?
14. How is tetany caused?

SECTION – III

III. Answer any Three questions. Question NO. is 19 Compulsory

3x3=9

15. Why elephants and other wild animals are entering into human living area?
16. List three features that characterise Bony Fishes.
17. How do Earthworm breathe?
18. What are the advantages of artificial insemination.
19. Differentiate between probiotics and pathogenic bacteria.

SECTION – IV

IV. Answer in Detail.

2x5=10

20. ABO blood Groups - Discuss.

(OR)

Explain the Absorption and assimilation of proteins, carbohydrates & fats.

21. Comment on the structure of thyroid Gland and its functions.

(OR)

Why the limbic system is called the emotional brain? Name the parts of it.



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